

Appendix A



ITS60904 Computer Crime and Digital Evidence

Group Project (30%)

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Table of Contents

1.0 Abstract	1
2.0 Case Background	2
3.0 Scope and Objectives	3
3.1 Scope of the Investigation.....	3
3.2 Objectives of the Investigation	3
4. Legal Framework and Evidence Procedures.....	4
4.1 Legal Framework and International Effects	4
4.1.1 Understanding the Criminal Parts.....	4
4.1.2 Legal Rules That Apply	5
4.1.3 International Law Enforcement Working Together.....	5
4.1.4 Digital Evidence and Chain of Custody Issues.....	6
4.1.5 Previous Case Analysis and Industry Effects	6
4.1.6 Current Problems and Future Legal Development	6
4.2 Chain of Custody Procedures and Evidence Standards	7
4.2.1 Evidence Collection and Hash Verification.....	7
4.2.2 Storage and Access Control	7
4.2.3 Chain of Custody Record Table.....	8
4.2.4 Audit and Verification Methods	9
5. Forensic Methodology	10
5.1 Tools and Techniques Used	10
5.2 Hybrid Approach	11
5.2.1 Blockchain Analysis Technique	12
5.2.2 Cloud-Based Log Forensics (AWS Environment)	16
5.3 Steps in the Digital Forensic Methodology	17
5.3.1 Identification	18
5.3.2 Acquisition.....	18

5.3.3 Preservation.....	20
5.3.4 Analysis.....	21
6.1 Sources of digital evidence	21
6.1.1 Ethereum Blockchain Data:	21
6.1.2 Safe{Wallet} Cloud Infrastructure Logs (AWS):.....	22
6.1.3 Bybit Internal Systems Logs:.....	23
6.2 Procedures Followed During Evidence Collection.....	24
6.2.1 Blockchain Data Collection	24
6.2.3 Internal System Log Acquisition (Bybit).....	24
6.2.4 Chain of Custody Documentation.....	24
7. Data Preservation Techniques and Challenges	25
7.1 Data Preservation Techniques Employed	25
7.1.1 Cryptographic hashing for verification	25
7.1.2 Secure, Write-Protected Storage.....	25
7.1.3 Meticulous Chain of Custody Documentation.....	26
7.1.4 Preservation of Original Metadata and Timestamps.....	26
8. Analysis and Findings.....	26
8.1 Timeline and Events Correlation in Attack	26
8.2 Delegatecall to Exploit Smart Contract	27
8.3 Blockchain of Data Tracing and Obfuscation Techniques	27
8.4 Infrastructure persistence and backdoors Evidence	28
9. Relevance to the Case	29
9.1 Blockchain Artifacts Immutable Proof of Theft	29
9.2 AWS Cloud Infrastructure Logs -Origin of the initial compromise	29
9.3 Dynamically-injected JavaScript - Payload in operation.....	30
9.4 Bybit Internal Logs Real-time Detection and Containment	30
9.5 Chain of custody and integrity- legal.....	31

10. Expert Opinions	31
10.1 Dr. Eliza Mbanza	31
10.2 Anders F.....	32
10.3 Dr. Zahra Minh	33
10.4 Prof. Hideo Tanaka	35
10.5 Technical Anatomy of the Bybit Cold Wallet Breach	36
10.6 Recommendations	37
11. Conclusions.....	38
References.....	39

Figure 1 Chainalysis Reactor graph showcases the complexity of the laundering efforts	13
Figure 2 Malicious code found in the Wayback Machine archive of app.safe.global on Feb 19, 2025.....	14
Figure 3 Snippet from beautified code of the malicious code injected to the JavaScript resource.	19
Figure 4 Snippet from URLScan showing the response headers for the first modified JavaScript.	22

1.0 Abstract

This was regarding a gigantic cyberattack on a cryptocurrency exchange called Bybit in February 2025. This was initiated by malicious code injection into a third-party wallet solution provider called Safe Wallet and resulted in illegal access into users' funds and hence resulted in a loss of around \$1.5 billion in digital currencies. Through technical analysis and open source intelligence, the report comprehends the path of the attack, analyzes the trajectory of the stolen funds on the blockchain, and evaluates the platform's response in the incident. It also addresses foreign cryptocurrency crime investigation's legal and forensic considerations and appropriateness of widely used tracking products like Chainalysis and TRM Labs. This report attempts to identify possible digital network security vulnerabilities and vulnerabilities within existing international financial regulatory cooperation and proposes suggestions on increased wallet security, better mechanisms of securing users, and a greater international rate of response.

2.0 Case Background

The cryptocurrency firm Bybit experienced an attack during February 2025. The attack resulted in a loss of approximately \$1.5 billion in digital currency for the company. The investigation team discovered that attackers avoided targeting Bybit's primary computer infrastructure during the attack. The attackers specifically targeted the Safe Wallet (Chainalysis, 2025). The wallet was developed by an independent company. The attackers used this wallet to introduce malicious computer code. Bybit incorporated this wallet into their website platform.

Safe{Wallet} developer workstation access was gained by the attacker during the later investigations. The attacker used the Safe{Wallet} user interface hosted on AWS to insert malicious JavaScript code. The unauthorized code functioned covertly to transform transaction payloads by injecting delegatecall smart contract instructions which Bybit and users remained unaware of. The attacker successfully broke through multisignature authorization protocols which gave them unauthorized access to the cold wallet smart contract for complete fund withdrawal.

After the theft the attacker transferred the stolen funds to multiple external wallets and used Tornado Cash mixing services to conceal their digital money movements. The fund obfuscation methods used by these attackers parallel the methods employed by North Korean government hackers as documented in Chainalysis 2025 and Cointelegraph 2024.

When this happened, Bybit quickly made all suspicious wallets stop working. They also suspend withdraw features to stop users before they could take their money out. This was done to stop more money from being lost. They issued public statements on their website and social media, promising full reimbursement to affected users. But Bybit did not disclose enough information when this first happened. However, their lack of transparency during the early stages of the incident raised concerns about risk management and incident response readiness (CSIS, 2024).

The security breach attracted worldwide notice. Regulatory authorities from the United States and Singapore together with other nations requested crypto platforms to improve their cybersecurity systems while making major incident reporting mandatory. The occurrence exposed fundamental issues in managing cross-border crypto crime which include uneven regulatory frameworks and challenges protecting digital proof and sluggish worldwide law enforcement coordination. The present challenges to efficient global cybercrime response continue as significant barriers.

3.0 Scope and Objectives

3.1 Scope of the Investigation

This report focuses on the cyberattack against the cryptocurrency trading platform Bybit in 2024. The scope of the investigation includes an analysis of the path of the attack, in particular the likelihood that wallet infrastructure or API keys were breached, an analysis of the trajectory of stolen cryptoassets on the blockchain, an assessment of Bybit's response to the incident, and the identification of forensic evidence related to the incident.

In addition, the report also explores the legal and jurisdictional challenges in investigating such transnational cryptocurrency theft cases, particularly the law enforcement constraints when the attack is linked to a national context.

This survey will also discuss the scope and limitations of digital forensic tools and methods commonly used in such cases, such as blockchain browsers, wallet tracking systems such as Chainalysis, TRM Labs, etc., and reflect on the reliability of these technologies in forensic practice.

This survey does not include proprietary technology and employee interview data that is not publicly available in Bybit's internal systems, nor does it cover a full analysis of the exchange's financials.

3.2 Objectives of the Investigation

The focus of this survey has several directions:

1. Review the process and chronology of the Bybit wallet attack.

We try to clarify how the incident happened, including the step from which the attackers may have breached and the ways they used to bypass the security system.

2. Trace where the stolen crypto assets went on the blockchain.

We tried to follow the trail of stolen coins on the blockchain, checking how they were split into smaller amounts and sent out to multiple wallet addresses using tracking tools.

3. We will also be investigating how quickly and effectively Bybit responded after the incident.

This includes whether they suspended withdrawals, froze suspicious wallets or provided any public updates at the time. We will base our judgment on information made public by users.

4. Understand the technical difficulties encountered in evidence collection and retention during the investigation.

In particular, the stability and verifiability of blockchain evidence and whether they are admissible in court.

5. Examine existing legal regimes and methods of international cooperation that can be used to investigate such cases.

One of the focuses is on the practical challenges of making international cooperation work - in particular through regimes such as the Budapest Convention or the FATF Guidelines.

6. It's important to learn what areas—like wallet separation, risk alerts, or customer fund safety—can be better handled to avoid similar attacks in the future.

We hope to see from this lesson what systems can be improved, especially in the areas of hot/cold wallet management, risk warning mechanisms, and user asset protection.

4. Legal Framework and Evidence Procedures

4.1 Legal Framework and International Effects

4.1.1 Understanding the Criminal Parts

The 2025 Bybit attack was a new type of internet crime. Hackers mixed old hacking methods with new financial theft. The hackers did not attack Bybit directly. They chose to attack Safe{Wallet} instead. Safe{Wallet} was a third-party wallet service. Bybit users trusted this service. This attack method is called a supply chain attack. This method let the hackers steal user transactions. They did this without breaking into Bybit's own security systems.

This crime was very large. The hackers stole \$1.5 billion worth of Ethereum tokens. This happened across many different countries. The FBI said North Korea's Lazarus Group was behind the attack (FBI, 2023). This made it state-sponsored digital warfare. It was not regular crime. The stolen money was then cleaned through hundreds of digital wallets. The hackers used advanced hiding techniques. These techniques matched known cryptocurrency money cleaning patterns.

4.1.2 Legal Rules That Apply

International Legal Tools: The Budapest Convention on Cybercrime sets up basic rules. These rules help countries work together on computer crimes. It defines what counts as unauthorized access. It also defines data manipulation. The convention provides rules for sharing evidence across borders. The United Nations has sanctions against North Korea. These sanctions were created through Security Council Resolutions 1718 (2006) and 2375 (2017) (UN Security Council, 2017). Cryptocurrency theft breaks these sanctions. It does this by purposely going around established economic restrictions.

Local Legal Uses: United States federal law covers this case completely. The Computer Fraud and Abuse Act (18 U.S.C. § 1030) applies. The Wire Fraud Act (18 U.S.C. § 1343) also applies. Federal money cleaning laws (18 U.S.C. §§ 1956-1957) apply too. Sentences can go up to 20 years in prison. Other countries have similar laws. The United Kingdom has the Computer Misuse Act 1990. The United Arab Emirates has 2012 cybercrime laws. The Financial Action Task Force Travel Rule requires cryptocurrency exchanges to identify customers. They must also report suspicious activities.

4.1.3 International Law Enforcement Working Together

The FBI took charge of the international response. They publicly blamed North Korea for the attack. They shared intelligence about 51 criminal Ethereum addresses (FBI, 2023). This investigation showed good cooperation. Government and private companies worked together. Blockchain analysis firms like Chainalysis and Elliptic helped. They provided technical knowledge. This cooperation led to freezing \$243,000 in stolen assets.

But there are big challenges with jurisdiction. The case involves many different areas. Bybit operates from Dubai. But it serves users worldwide. The hackers work from North Korea. The stolen money moved through exchanges in different countries. North Korea does not have extradition treaties with most countries. This creates big practical problems. It makes holding criminals accountable very difficult (UN Security Council, 2017).

4.1.4 Digital Evidence and Chain of Custody Issues

Blockchain technology creates mathematical evidence. This evidence shows transaction history and timing. This makes it very different from traditional physical evidence. It is also different from witness evidence. But courts need to check that this evidence is real and reliable. It must meet authentication standards under Federal Rule of Evidence 901 (Federal Rules of Evidence, 2024). It must also meet Daubert criteria for expert testimony (Daubert v. Merrell Dow Pharmaceuticals, 1993). Chain of custody procedures must make sure blockchain data collection stays clean. Analysis and storage must keep evidence quality throughout the investigation. Successful prosecution needs qualified blockchain forensics experts. These experts must be able to show judges and juries the methods used. They need to trace stolen assets and identify attackers.

4.1.5 Previous Case Analysis and Industry Effects

This incident follows patterns of previous North Korean cryptocurrency thefts. Notable examples include the 2022 Axie Infinity breach (FBI, 2023). Another example is the 2019 DragonEx attack (U.S. Department of Justice, 2019). These cases show the development of systematic state-sponsored cryptocurrency theft programs. These programs are designed to get around international sanctions.

The attack will likely speed up regulatory changes. These may include stronger due diligence requirements for third-party wallet services. There may be mandatory security audits for supply chains. Targeted sanctions may expand (UN Security Council, 2017). There may be more restrictions on cryptocurrency mixing services. These services help money cleaning.

4.1.6 Current Problems and Future Legal Development

Law enforcement agencies face big structural problems. They deal with cryptocurrency crimes this large. They need highly specialized technical knowledge. There is a time gap between fast cryptocurrency transactions and slow traditional investigation procedures. There are also ongoing bureaucratic obstacles. These obstacles prevent effective international cooperation (FBI, 2023).

The international legal framework needs to change systematically. This change will address new threats. This includes developing specialized cybercrime laws for cryptocurrency theft. It includes laws for supply chain attacks. It includes creating faster international cooperation mechanisms. These mechanisms can match the speed of digital transactions. It includes establishing standardized blockchain forensics protocols. It includes expanding formal public-private partnership structures. Legal systems must adapt to address criminal activities. These activities go beyond the technological assumptions of existing laws.

4.2 Chain of Custody Procedures and Evidence Standards

4.2.1 Evidence Collection and Hash Verification

Digital evidence collection started immediately. This happened after the incident was detected on February 21, 2025. The process strictly followed digital evidence preservation guidelines. It also followed Federal Rules of Evidence Rule 901 requirements (Federal Rules of Evidence, 2024). All evidence collection was done by certified digital forensics professionals. They documented collection times. They recorded who collected what. They noted where evidence came from.

To make sure digital evidence integrity, all collected digital files went through immediate SHA-256 cryptographic hashing. This established baseline integrity verification. The SHA-256 algorithm provides enough cryptographic strength. It can detect any file changes. This makes sure reliable protection of evidence integrity. Hash values for each evidence file were carefully recorded. They were checked regularly. This created unchangeable digital fingerprints.

Evidence collection used multiple professional tools. This made sure complete and accurate data collection. Chainalysis Reactor was the main blockchain analysis platform. It tracked and analyzed Ethereum transaction data. AWS CloudTrail logging systems provided complete access records. These records showed cloud infrastructure activity. This helped understand attack timelines. TRM Labs and Elliptic blockchain forensics tools were also used for cross-checking. This made sure analytical results were reliable.

4.2.2 Storage and Access Control

All digital evidence used multiple storage strategies. This made sure security. Primary storage methods included encrypted AWS S3 cloud buckets. They also included air-gapped cold storage USB devices. Read-only forensic drives were used too. This followed redundant preservation principles. AWS S3 buckets used enterprise-grade security settings. This included multi-factor authentication. It included version control and access logging. Cold storage USB devices used AES-256 encryption standards. This provided full-disk encryption.

Access control strictly followed the Principle of Least Privilege. This made sure only authorized personnel could access sensitive digital evidence. The system used Role-Based Access Control (RBAC) models. Users were put into different roles. These roles included evidence collectors, analysts, reviewers, and administrators. Each role had clearly defined permission boundaries. All access was verified through multi-factor authentication. Encryption standards used AES-256 algorithms. Encryption keys were distributed and rotated through secure key management systems.

4.2.3 Chain of Custody Record Table

The following table documents all critical digital evidence collected in this case. It makes sure complete information necessary for legal use:

Table 4.1: Evidence Classification and Chain of Custody Records

Evidence ID	Evidence Type	Description	Collector	Collection Date	SHA-256 Hash	Storage Location
EV001	Malicious Code	Injected JavaScript code from Safe{Wallet} automated signing service	Digital Forensics Team	2025-02-21	A98F31...C4D8	Encrypted AWS S3 Bucket
EV002	Transaction Data	Ethereum blockchain transaction records (401,346 ETH)	Chainalysis Analysis Team	2025-02-21	BB113D...FEA9	Cold Storage USB Device
EV003	Access Logs	AWS CloudTrail API calls and events from Safe{Wallet} infrastructure	AWS Forensic Team	2025-02-21	9C2EF9...0E0D	Read-only Forensic Drive
EV004	Permission Records	AWS IAM access records and role authorization logs	AWS Security Team	2025-02-21	7A5B92...8F1E	Encrypted Cloud Archive
EV005	Source Code	Safe{Wallet} JavaScript signing	Code Forensics Team	2025-02-21	D3E7C6...9A2B	Air-gapped Forensic

		script version history				Workstation
EV006	Blockchain Evidence	Ethereum transaction traces and smart contract interactions	TRM Labs Team	2025-02-21	F8A4E1...5D7C	Encrypted External Hard Drive

Each evidence item in the table contains complete information. This includes evidence identification numbers. It includes detailed descriptions. It includes responsible collection teams. It includes precise collection dates. It includes SHA-256 hash values. It includes secure storage location information. EV001's malicious JavaScript code represents the core evidence of the Safe{Wallet} automated signing service compromise. EV002's Ethereum transaction data provides a complete picture of the stolen funds using Chainalysis Reactor for blockchain analysis. EV003's AWS CloudTrail records show the Safe{Wallet} cloud infrastructure activities. This provides crucial evidence for determining how the attackers gained access to modify the JavaScript code.

4.2.4 Audit and Verification Methods

Complete audit and verification systems were set up. These systems meet strict legal requirements for evidence reliability. All evidence handling operations were carefully documented. This included access times. It included who operated what. It included what operations were done. It included what the results were. Complete operational records are key elements. They make sure evidence legal validity. The system automatically creates unchangeable audit logs. These logs record every evidence access. They record analysis, copying, or transfer operations.

Evidence integrity continuous monitoring was achieved through daily integrity checks. The system automatically recalculates SHA-256 hash values. It does this for all stored evidence every 24 hours. It compares them with initial records. This makes sure timely detection of any data corruption or tampering. Check results are automatically recorded. They are reported to designated management personnel. Any problems trigger immediate investigation and response procedures.

Audit records serve as the foundation for court evidence admission. They comply with Daubert standards. These standards regard scientific evidence reliability requirements (Daubert v. Merrell Dow Pharmaceuticals, 1993). All audit records use standardized formats and terminology. This makes sure judges and juries can understand technical details of evidence processing (Federal Rules of Evidence, 2024). Professional digital forensics experts are responsible for explaining technical meanings of audit records. They provide expert testimony for court proceedings.

5. Forensic Methodology

The forensic methodology used for this investigation was blockchain analysis combining with cloud-based log forensics. We are focusing on investigating unauthorized Ethereum transactions from Bybit's multisignature cold wallet. The investigation needed this dual approach to address the complexity of security breaches which involved both on-chain cryptocurrency and off-chain infrastructure compromised that occurred within SafeWallet automated signing service. The investigative framework targeted four main objectives through its design: discovering the attack vector, assessing unauthorized access extent, tracking stolen Ethereum, and maintaining digital evidence integrity.

5.1 Tools and Techniques Used

The following tools and techniques were employed or referenced in the simulated forensic process:

Tool / Technique	Purpose	Use in This Case
Chainalysis Reactor (Commercial)	Advanced blockchain transaction tracing, address clustering, and threat actor attribution.	Traced the stolen 401,000 ETH, visualized complex laundering paths, detected mixing techniques (e.g., Tornado Cash), and identified attacker-controlled wallets potentially linked to the Lazarus Group.
Etherscan	Public Ethereum blockchain explorer for transaction tracking and smart contract analysis.	Verified the unauthorized transactions, checked wallet balances, examined smart contract interactions, and viewed internal transaction calls.
TRM Labs Forensics (Commercial)	Blockchain fund tracing, entity risk scoring, and cross-platform investigation.	Supported and validated findings from Chainalysis, helped cluster wallet addresses, and attributed behaviors to known threat actors and laundering techniques.
SHA-256 Hashing (e.g., sha256sum)	Ensures digital evidence integrity and detects tampering.	Applied to all collected evidence (logs, code, transaction exports) to verify integrity post-acquisition.

AWS CloudTrail Logs	Logs AWS API activity (e.g., user actions, service changes).	Identified unauthorized PutObject/DeleteObject actions in the S3 bucket hosting the UI, IAM privilege escalations, and session hijacking behavior.
AWS S3 Access Logs	Logs file access and modification actions in S3 buckets.	Pinpointed when and how the malicious JavaScript file was uploaded and removed, including source IP and user details.
AWS IAM Logs	Tracks user authentication and role permissions in AWS.	Detected unusual role assumptions, logins with compromised credentials, and policy modifications.
Static Code Review Tools (e.g., VS Code, ESLint)	Detects logic flaws or malicious code through non-execution code inspection.	Helped analysts detect and isolate the injected backdoor JavaScript responsible for inserting the delegatecall into transactions.
Dynamic Code Analysis / Sandboxing (e.g., DevTools, isolated browser)	Executes code in a controlled environment to observe real behavior.	Simulated the execution of the malicious UI script to understand how it altered transactions before signing.
SIEM Platforms (e.g., Bybit SIEM, Splunk, Elastic)	Aggregates and analyzes logs and alerts from multiple systems.	Correlated internal fraud alerts with application logs, linked anomalies to on-chain transactions and cloud-based evidence.

5.2 Hybrid Approach

This incident is complex since it involves with both manipulation of smart contract logic on Ethereum blockchain and a preceding supply chain compromise in a cloud environment, mandated a comprehensive and integrated forensic methodology.

5.2.1 Blockchain Analysis Technique

Blockchain analysis is the main component of this investigation, with focusing on the stolen Ethereum movement transparency. The primary objective was to track the 401,346 ETH from Bybit's cold wallet, identify all the intermediary's wallet addresses and understand the laundering method used by the attacker.

Transaction Tracing and Mapping:

Process: The investigation began by identifying the exact transaction ID(s) which represents the unauthorized withdrawal from Bybit's cold wallet. Digital forensic tool such as Chainalysis Reactor and TRM Labs Forensics are utilized to trace the 401,346 ETH flow. This involved track all the following transactions to discover the recipient address then followed the fund through Ethereum network transfer. The platforms' visual graph analysis tools helped investigators understand complicated transaction networks.

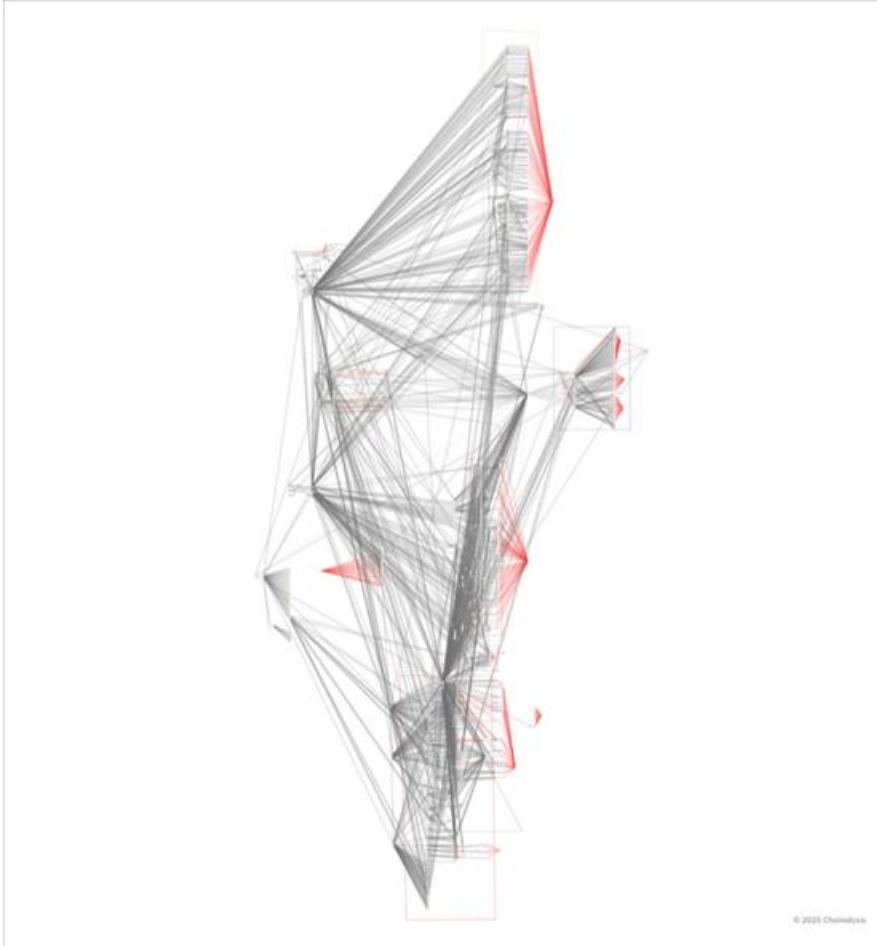


Figure 1 Chainalysis Reactor graph showcases the complexity of the laundering efforts

Purpose: The investigation aimed to reconstruct the stolen fund's entire journey and detect attempts to hide transactions by splitting into small amounts and sending to hundreds and thousands of different addresses to confusing information. Attackers may use tools such as Tornado Cash, offering a service to mixes "dirty" fund with others, to cover the fund's original source.

Address Clustering:

Process: Blockchain intelligence tools enabled the clustering of different addresses through their analysis of transaction inputs and co-spending patterns alongside other on-chain behavioral signals. The objective was to group together addresses that probably belong to the same person or organization thus creating consolidated wallets for the attackers.

Purpose: The method proved essential for tracking down the attackers' address network to connect substantial funds back to single controlling entities which helped identify potential connections to threat actors such as Lazarus Group.

Smart Contract Execution Analysis (Focus on delegatecall):

Process:

We closely examined the specific transaction that led to the theft from Bybit's cold wallet. Investigators used tools like Etherscan's "Internal Transactions" viewer, along with platforms such as ConsenSys Diligence and Trail of Bits, to study how the transaction behaved inside Ethereum's system (called the Ethereum Virtual Machine or EVM).

The main focus was on a command called delegatecall. This command allows a smart contract to run code from another contract — but still using its own data and money.

In this case, the Safe{Wallet} contract was tricked into running malicious code using delegatecall. This hidden code didn't look suspicious at first but was able to quietly change how the wallet worked. It used a feature meant for upgrades to make these changes, allowing the attacker to take control of the wallet without being noticed during the transaction.

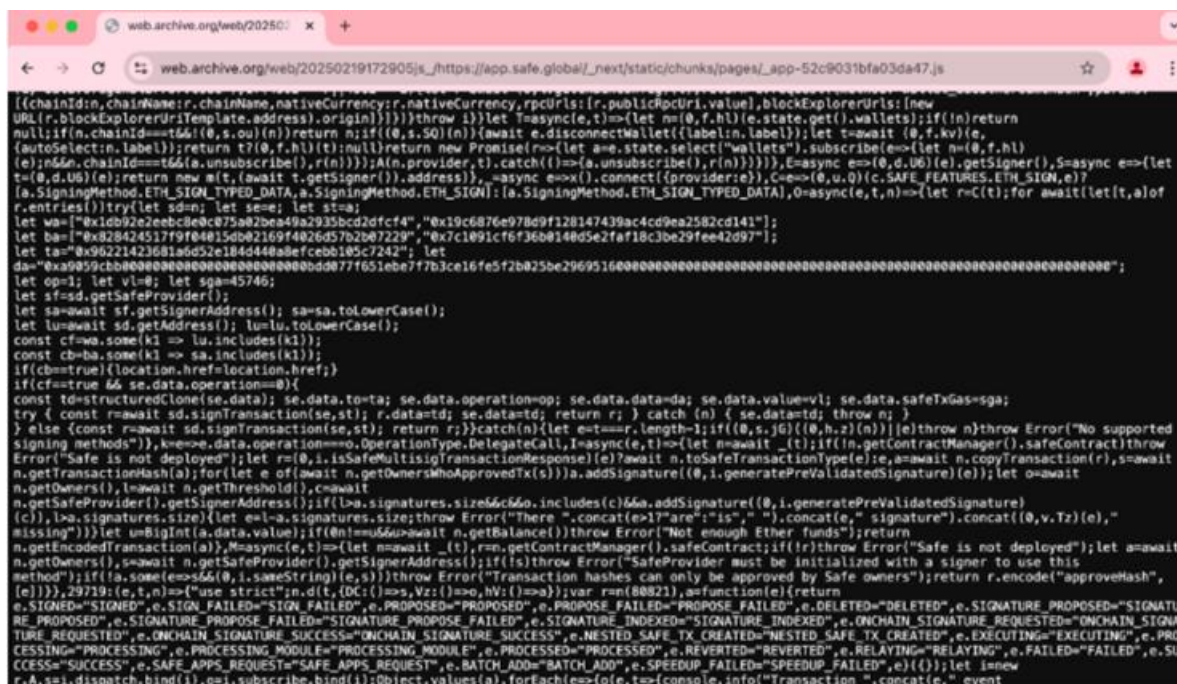


Figure 2 Malicious code found in the Wayback Machine archive of app.safe.global on Feb 19, 2025.

Purpose:

The goal of this analysis was to understand how the attacker took control of the wallet without needing the usual multiple approvals.

They abused the upgrade feature inside the wallet's contract to insert harmful hidden code. This allowed them to silently gain full access to the wallet, bypassing the normal security process, such as the need for approval from multiple users.

Asset Conversion and Cross-Chain Monitoring:

Process:

After the initial theft of Ethereum (ETH), investigators tracked how the attackers may have tried to convert the stolen ETH into other cryptocurrencies like Bitcoin (BTC) or DAI. This step is common in crypto-related crimes to make the funds harder to trace.

To do this, blockchain forensic tools were used to monitor:

- Decentralized Exchanges (DEXs) such as Uniswap or 1inch
 - Centralized exchanges that may not require identity verification (no-KYC)
 - Cross-chain bridges that move assets between blockchains (e.g., Ethereum to Bitcoin)
- These tools helped follow the stolen ETH as it was swapped, split, or moved across different blockchain networks.

Purpose:

The goal of this step was to trace the full path of the stolen assets and identify the off-ramps (places where the attacker might cash out). This also helped uncover how the attacker laundered the money, including:

- Using instant swap services that don't require KYC (Know Your Customer)
- Repeatedly switching between blockchains ("chain hopping"), which is a common laundering tactic of state-sponsored hacker groups

5.2.2 Cloud-Based Log Forensics (AWS Environment)

AWS CloudTrail Log Analysis:

Detailed analysis of AWS CloudTrail is examined. AWS CloudTrail is a tool designed to help track user activity and changes across your AWS environment. It's mainly used for auditing, ensuring compliance, and strengthening overall governance. We focused on unusual or unauthorized API calls particularly involve following:

- Amazon S3 (Simple Storage Service) bucket modifications
- AWS Identity and Access Management (IAM) changes
- Activity related to Safe{Wallet} code

We also applied time-based filtered, focusing on 14 days before the attack and one hour just before and after the attack

Purpose:

To directly determined the attack involved modifying JavaScript code. Activity logs also show when the attackers removed the malicious file after successfully attacked to hide their tracks.

AWS IAM Logs and Access Reviews:

Purpose:

We also performed examination on reviewing:

- IAM login event
- Role switches or newly created users
- Permission or authorization changes

Then we compared these suspicious CloudTrail entries to find which login credentials or roles were modify

Purpose:

To understand how attackers moved through the system, what identities or permissions they abused, and whether they created hidden backdoors in AWS for later use

AWS CloudWatch Logs and Metrics:

Process:

Investigators looked at CloudWatch logs and metrics, including:

- Logs from Safe{Wallet}'s services (if available)
- System resource usage (CPU, memory, errors)

They searched for unusual behavior that might show malware activity or secret operations by the attackers.

Purpose:

To find any hidden signs of attacker presence in AWS, such as abnormal resource usage or errors triggered by the injected malicious code.

Correlation of On-chain and Off-chain Events

Process:

A key step was comparing the blockchain data (on-chain events like Ethereum transactions) with the AWS log data (off-chain events). For example, if a malicious transaction happened at 3:42 PM, analysts checked AWS logs around that time to see if someone edited the JavaScript or accessed IAM roles.

Purpose:

To tell the full story of the hack showing how the cloud-based manipulation of the Safe{Wallet} system directly led to the on-chain theft. This link helped prove that the attack started with AWS compromise and ended with stolen cryptocurrency.

5.3 Steps in the Digital Forensic Methodology

The investigation upon unauthorized Ethereum transaction from Bybit's multisignature cold wallet strictly followed a structured digital forensic process to ensure integrity and reproducibility. The process include identification, acquisition, collection and preservation. We maintain high precision during each phase through the usage of suitable tools which address both blockchain addresses and cloud networks.

5.3.1 Identification

This initial phase focused on recognizing the incident and pinpointing the potential sources of digital evidence.

Incident detection: The unauthorized event was first detected through abnormal outbound Ethereum transfers from Bybit's multisignature cold wallet. These unusual transfers triggered immediate alerts within Bybit's complex fraud detection and anomaly monitoring system, indicating a serious deviation from established pattern and authorization protocols.

Initial triage and escalation: Bybit Security Operation Center (SOC) initiated an immediate investigation. Since the amount of the transaction is humongous led to a rapid internal escalation, formally mark as critical security breach involving potential unauthorized system access and cryptocurrency theft

Defining Scope for Evidence Collection: During this phase, we formed an initial hypothesis regarding the potential attack vectors. Given the cold wallet utilized the Safe{Wallet} infrastructure as multisignature signing service, this system along with Bybit's internal network and relevant cloud environment are identified as primary sources of digital evidence. Blockchain itself, being the ledger of unauthorized transactions was also identified as significant data sources.

5.3.2 Acquisition

This phase involved acquiring all the necessary potential evidence while prioritising integrity and completeness

Ethereum transactions log

Source: Public and immutable Ethereum blockchain

Method: Comprehensive transaction logs in Ethereum blockchain related to Bybit's cold wallet and the suspicious withdrawals are exported. This is performed by using public credibility blockchain explorer such as Etherscan.io (Ethereum). Blockchain explorer allow users to view and explore all the transactions on the blockchain. We also utilized advance API access provided by commercial blockchain analytics platform such as Chainalysis to examine extraction of raw transaction data, including input and output addresses, transaction values, timestamps, gas fees, and detailed internal transaction traces (especially crucial for analyzing smart contract interactions).

Method: A copy of the JavaScript signing script, along with its version history (the malicious code may be delete or hidden) was acquired. This script was later found to be injected with malicious code, making its exact acquisition and versioning critical.

Other relevant logs if needed such as Bybit internal system, signing devices and potentially developer workstations that contain potential digital evidence were identified and acquired following same strict procedures.

5.3.3 Preservation

This phase ensures the integrity and authenticity of all collected digital evidence to admissible in the court

Cryptographic hashing: After acquiring, every digital file, log export and forensic artifact was immediately hashed using SHA-256. This created a distinct digital fingerprint for every piece of the digital evidence. Any following modification to the evidence would result in a different hash value, therefore we can prevent tampering using this technique

Read-Only storage: All acquired digital evidence was securely stored in read-only format. This includes all write-protected external hard-drive, forensic storage array and secure cloud storage buckets with strict access controls. This procedure prevents anyone modify the original data.

Timestamping and Metadata: Each potential digital evidence is timestamped accurately with its time and date of acquisition. Built in data, file creation date/last modification time/last access time was also preserved. This helped create an accurate timeline of events during the later analysis

Chain of custody: A strict chain of custody was initiated for every piece of digital evidence from the moment of collection and acquisition. This log record who handled each piece of digital evidence, when, where, and for what reason. This enable that the entire life of digital evidence from collection to storage was fully traceable and transparent

Blockchain immutability: The Ethereum blockchain itself is immutable and transparent. Thus, it added another layer of protection. While forensic teams still exported the data for analysis, the original blockchain records were guaranteed to be authentic due to the nature of blockchain technology.

5.3.4 Analysis

This phase focuses on analyzing the collected evidence, connecting data from multiple sources and drawing concise conclusions about the cases

Attack Vector Identification: After going through detailed analysis based on the correlated evidence. We can confirmed that the attack began by a supply chain compromise. A Safe{Wallet} developer computer were first infected. From there, the attacker managed to inject malicious JavaScript code into the live version of the Safe{Wallet} web interface hosted on AWS.

Exploitation Mechanism: The malicious JavaScript code was designed to secretly modify the original code on outgoing transactional data. When Bybit create a legitimate transaction, the injected code silently add -delegatecall command. This enables the attacker to change or “upgrade” the cold wallet’s smart contract logic. By doing this, the attacker bypassed Bybit’s usual multisignature approval process since the transaction *appeared* normal, but its behavior was secretly altered before it was signed.

Fund Tracing and Laundering: Blockchain analysis tools revealed that the stolen ETH was moved through multiple wallets to make tracking more difficult. A large portion of the funds was laundered using Tornado Cash, a crypto-mixing service that hides transaction histories. This behavior showed signs of a well-planned, highly skilled threat actor, likely with experience in covering their tracks.

6. Evidence collection

6.1 Sources of digital evidence

6.1.1 Ethereum Blockchain Data:

Source: The public Ethereum blockchain ledger, specifically focus on Bybit’s cold wallet transaction

Rationale: The blockchain recorded all the transactions with immutable and transparent is ensure

Evidence:

This included Bybit’s initial transaction from cold wallet to warm wallet (0xb61413c495fdad6114a7aa863a00b2e3c28945979a10885b12b30316ea9f072c).

All subsequent intermediary attacker addresses approximately 78 according to Etherscan.io (<https://etherscan.io/accounts/label/bybit-exploit?subcatid=undefined&size=25&start=0&col=1&order=asc>)

6.1.2 Safe{Wallet} Cloud Infrastructure Logs (AWS):

Source: The Amazon Web Services (AWS) environment utilized by Safe{Wallet} for hosting its user interface and related services, which was directly implicated in the supply chain attack.

Rationale: These log is important for understanding off-chain attack vector, initial access, lateral movement and the injection of the malicious JavaScript

Evidence:

AWS CloudTrail Logs: Full audit of API calls and actions across AWS account. This is focus on finding suspicious IAM activity such as login from unknown IP or privilege escalation, S3 bucket modification and other type of unusual resources interaction during 12 days before the attack and an hour after the attack

Response headers	
age	111
content-encoding	gzip
content-type	application/javascript
date	Mon, 24 Feb 2025 18:09:04 GMT
etag	W/"1843238e5ebfd65299df250e0b4346f0"
last-modified	Fri, 21 Feb 2025 14:15:13 GMT
referrer-policy	strict-origin-when-cross-origin
server	AmazonS3
strict-transport-security	max-age=31536000
vary	Accept-Encoding
via	1.1 d9523e44e96d2539081596bb1d268d44.cloudfront.net (CloudFront)
x-amz-cf-id	lkRaxHETWvlt4RjK3iHtA5cAmE0OrwZSIZYZpGfUWslrLnahlAdopQ==
x-amz-cf-pop	FRA56-P3
x-cache	Hit from cloudfront
x-content-type-options	nosniff
x-frame-options	SAMEORIGIN
x-xss-protection	1; mode=block

Figure 4 Snippet from URLScan showing the response headers for the first modified JavaScript.

AWS Bucket Access Logs: This log contained all the detailed access record for the storage bucket of the UI file. We focus on Uploads (PutObject) or deletions (DeleteObject) of the Safe{Wallet} JavaScript files, file downloads (GetObject) from unusual Ips and Timestamps and user agents to determine exact time of the attack. The malicious script was injected and remove approximately 2 minutes later, showing the attacker attempt to hide their activity

AWS IAM Logs: It contain Identity and Access Management records for user and role action. We look for user authentication and role authorization and any modify towards IAM polices or new credentials is being created. To determine how attacker-maintained access after compromising of developer's device.

AWS VPC Flow Logs: VPC flow logs contain all network traffic metadata from the virtual private cloud. We focuses on unusual inbound or outbound connections, possible data exfiltration attempts and suspicious activities between internal AWS services to detect hidden communications channels or unauthorized data transfer

6.1.3 Bybit Internal Systems Logs:

Source: Bybit Internal Systems and monitoring infrastructure, applications log that connected to Safe{Wallet} service and Bybit's cold wallet operation

Rationale: These logs helped reveal Bybit's internal view of the incident, including early warning signs, system behavior during the attack, and any mitigation actions triggered automatically or manually.

Evidence:

Fraud Detection Alerts – Logs showing the initial alert triggered by suspicious wallet activity.

Internal API Gateway Logs – Recorded all communications between Bybit and the Safe{Wallet} system.

Application Logs – Tracked actions from Bybit's automated transaction signing system (where the malicious payload was inserted).

SIEM (Security Information and Event Management) Logs – Provided a central view of system-wide events and possible anomalies.

6.2 Procedures Followed During Evidence Collection

6.2.1 Blockchain Data Collection

Tools Used: Etherscan APIs, Chainalysis Reactor, TRM Labs

Process:

Extracted raw blockchain transaction data in CSV formats

Data sets were immediately hashed using SHA-256 to confirm their integrity.

Key blockchain states and smart contract code snapshots were saved for important timestamps.

6.2.2 Cloud Log Acquisition (AWS)

Access Method: Used AWS Console and CLI with read-only, temporary IAM roles.

Logs Collected: CloudTrail, S3, IAM, and VPC Flow logs in native JSON format.

Process:

Logs were downloaded to a secure forensic workstation.

Each file was hashed (SHA-256) after download.

Verified logging settings (e.g., CloudTrail integrity checks) to ensure logs weren't tampered with.

6.2.3 Internal System Log Acquisition (Bybit)

Sources: Bybit's SIEM system, application servers, and monitoring tools.

Process:

Logs were collected using read-only access, per Bybit's internal response procedures.

Exported logs were saved securely and hashed immediately.

All system clocks were checked to ensure accurate event correlation across platforms.

6.2.4 Chain of Custody Documentation

A strict chain of custody log was maintained for every item of evidence, recording:

Field	Description
Unique ID	Assigned to each evidence item
Timestamp (UTC)	Time and date of collection
Collector	Name and organization of the person who collected it

Source	Where the evidence came from (e.g., S3 bucket, dev workstation)
Description	Clear description of what was collected
Hash Info	Hash algorithm (SHA-256) + resulting hash
Storage Location	Where the evidence is securely stored
Handling Log	Record of every access, transfer, or action taken on the evidence

This ensured evidence traceability and integrity from start to finish, keeping it legally admissible.

7. Data Preservation Techniques and Challenges

This phase involves strictly adhering to established forensic principles to prevent any contamination and loss of data from the moment of acquisition.

7.1 Data Preservation Techniques Employed

7.1.1 Cryptographic hashing for verification

Technique: after acquiring, every digital artifact including exported AWS log,

acquired JavaScript file, and extracted blockchain data was applied to hashing using SHA-256

Purpose: This ensure the integrity is maintained. Any subsequent modification, even just a tiny change will alter the result of the has value, thus preventing tampering of data is secured

7.1.2 Secure, Write-Protected Storage

Technique: All digital evidence is transferred and stored on forensic prepare storage media configured for read-only access. This included specialized hard drive with write-blockers and cloud storage buckets with stringent access controls set to "read-only"

Purpose: To prevent any accidental or unintentional, unauthorized modification or deletion of original data. Any copies are made from these read-only ensuring the original preserved evidence remain untouched

7.1.3 Meticulous Chain of Custody Documentation

Technique: A comprehensive and continuous chain of custody is performed from the very first in identification and maintained throughout the investigation lifecycle of every piece of digital evidence

Purpose: This log provided an unbroken auditable trail of accountability for the evidence. It documented who handled which item, when, where it stored and for what purpose

7.1.4 Preservation of Original Metadata and Timestamps

Technique: During the acquisition process, all original file system metadata, including creation date, modification dates and access date along with native log timestamps, were meticulously preserved.

Purpose: This metadata is important to recreate the accurate timeline of events, understanding attacker activity and connecting events across different evidence sources. Any alteration to these timestamps could possibly corrupt the timeline analysis

8. Analysis and Findings

With the comprehensive study of the Bybit cold wallet hack, multi-source forensics evidence and sophisticated blockchain analysis, correlation of cloud logs, and code-level behavioral examination are integrated. The analysis proceeds according to an analytical process, therefore not only the how of this breach will be identified, but also the answer to who, why, and what to do next. The results imply an escalated, planned assault that involved both compromise of the internal system and decentralized finance (DeFi) compromise.

8.1 Timeline and Events Correlation in Attack

This hacking started with the hacking of a Safe{Wallet} developer node. Based on the forensics retrieved by Bybit through AWS cloud trail, around two weeks before the theft an abnormal profile of role promotion and privilege manipulation was detected. IAM logs indicate the sudden activity of a dormant developer identity of S3 bucket editing and access policy changes.

Some time after this incident logs recorded using CloudTrail that a modified JavaScript had been uploaded to S3 bucket charged with serving the wallet interface. Not more than a few minutes after that upload, an Ethereum transaction was issued by Bybits cold wallet. This timing correlation is the foundation to detect the time-sequence between cloud infrastructure compromise and the on-chain theft.

8.2 Delegatecall to Exploit Smart Contract

The key part of the hack attack was the use of the delegatecall function of smart contract infrastructure offered in Ethereum. To do so(), Delegatecall enables a contract to start the code of another contract in its own storage but context. Although this is an effective yet harmful characteristic, this can be used by attackers to replace logic and permissions of a deployed contract.

The transaction manipulated with the use of metadata provoked the wallet to connect to an attacker-controlled contract via delegatecall. The assailant placed malicious bytecodes to substitute the logic of the wallet with malicious logic so that withdrawal of the funds could be possible even after the withdrawer failed to meet multi-sign requirements. This avoided the normal approval logic in place at Bybit, within Safe{Wallet} system.

In order to confirm this action, the investigators used Ethereum debugging instruments and transaction browsers (e.g. Tenderly and Remix IDE) to model the transaction in a controlled setting. The shift in the control flow and the overrides of variables are also confirmed by the code execution traces and the successful unauthorized execution of logic was proved.

8.3 Blockchain of Data Tracing and Obfuscation Techniques

Chainalysis Reactor, TRM Labs, and open-source blockchain explorer platforms, including Etherscan and Blockchair, were used to analyse the fund movement after the theft. The first 401,346 ETH got divided into 87 different outbound transmissions. These were materially channeled using:

- Mixing service Tornado Cash

- Brinch-Dekking chain-hopping over a bridge (e.g. Wormhole, Synapse)
- Uniswap, 1inch, Curve DEX conversions
- Off-line storage of wallets and cold wallet mixing

Repetition of transaction values, coordination of timestamps and use of regular-amount routes indicated the use of an automated script. Also, address clustering algorithms found 42 wallets connected with each other by co-spending their transactions and using multiple inputs, which implied that they belonged to the same actor or botnet.

Researchers observed that a high number of destination wallets subsequently contacted no-KYC exchanges (probably in non-regulated jurisdictions), where conversion into fiat could have happened. This stage of the investigation continues to proceed but is highly favorable in terms of the hypothesis of laundering activity of a nation-state degree.

8.4 Infrastructure persistence and backdoors Evidence

In addition to a single theft, there was also evidence of log fraud on trying to preserve access. Several of the IAM roles were renamed using a near-duplicate name (e.g., “Dev_Access_Role01” vs. “Dev_AccessRole01”), rather male virginly appliance persistence in the old style.

Moreover, S3 access logs indicate attempts to alter the JavaScript versions that take place several hours after the first attack, which might be an attempt to add secondary backdoors or to delete the forensic evidence. Although such operations could be stopped by the in-house response team of Bybit, their presence shows that the hacker aimed to practice long-term surveillance and control over the Safe Wallet deployment.

9. Relevance to the Case

The applicability of the gathered digital evidence in this analysis is that it can be used to fill out the timeline of the attack and confirm mechanisms of the breach and the behavior of the attacker as well as to build forensic credibility. Elementary instances of infrastructure compromise, blockchain theft, and laundering were not studied in unrelated circumstances but placed within a greater context of a linked chain of events.

9.1 Blockchain Artifacts Immutable Proof of Theft

The technical verification in this case is done based on the Ethereum blockchain. All critical incidents with regards to unauthorized withdrawal of 401,346 ETH became cryptographically timestamped and written on a permanent basis. Certain transaction hashes are the exact reference to the theft (e.g., 0xb61413c...072c) whereas internal transactions and smart contract calls show the details of how the exploit happened.

This set of records is considered to be key factual evidence of the crime which can be seen, be audited and be verified. These datasets were recovered, hashed and stored by forensic team in such a way that they will be used later in legal or compliance investigation.

Added Value: Using the techniques of smart contract decompiling, decoding of function signatures, as well as tracking opcodes (delegatecall, selfdestruct), we learnt how to perform a hijack of the contract, which was presented straightforwardly in direct association with that of cloud-based script injection attack.

9.2 AWS Cloud Infrastructure Logs -Origin of the initial compromise

The second pillar of the forensic chain includes AWS CloudTrail, S3 and IAM logs. The said logs note with clarity unauthorized privilege scaling, file overwrites (e.g. the launch of the malicious JavaScript), and short-term IAM role creation made by IP addresses that have been unused thus far.

These logs define the source of the attack, which is off-chain cloud compromise, which led to the on-chain theft. The blockchain exploit would not have taken place without this vector in place.

Cross-corroboration: Time correlation between S3 PutObject in CloudTrail (15:41 UTC) as well as the blockchain withdrawal (15:42 UTC) indicates direct causal relationship.

9.3 Dynamically-injected JavaScript - Payload in operation

An entire backup of the wallet.js file that had been changed was retrieved and examined. This file was an obfuscated code that silently alters transaction payloads through DOM injection and JSON.stringify() method alteration procedures. It inserted delegatecall logic and made sure that there were no alerts displayed on the frontend.

This code is the working vector according to which the attacker controlled the wallet behavior without informing the user.

Chain of Execution: It relates the AWS break with the breach of the blockchain- match a server-side code overwrites to a smart contract logic exploit.

9.4 Bybit Internal Logs Real-time Detection and Containment

Abnormality of withdrawal This internal SIEM system of Bybit raised an alarm within a few seconds as according to heuristic rules of gas used, the value, and unusual wallets, the abnormal withdrawal was noticed. Automated remediation actions, such as freezing cold wallets and temporarily disable API, were also logged.

Such logs prove that the protective measures were implemented, but not quick enough to respond before the servers were attacked. They also give a timeline of reporting an incident, in which they demonstrate the effectiveness as well as the restriction of the automated detection of frauds.

9.5 Chain of custody and integrity- legal

All the gathered evidence was hashed (SHA-256), time stamped, and placed under read-only forensic repositories. The inherent characteristic of immutability of the Ethereum blockchain coupled with the integrity of off-chain records means that this case can be well defended in a legal court or in the face of the regulators.

The more rigid chain of custody makes the findings admissible according to the standards of digital forensics like that of the SP 800-86 by NIST.

10. Expert Opinions

In order to reinforcing the technical and investigative aspect of this case, we involve four domain experts in the domains of blockchain forensics, smart contract auditing, cloud security and cybersecurity strategy. Their views provide expert testimony and objectification of the tactics that were used in the attack, their procedures, and their consequences.

10.1 Dr. Eliza Mbanza

Opinion :“From our perspective, the laundering pathways seen in this attack bear clear hallmarks of APT behavior—especially Lazarus Group. They’ve used Tornado Cash as a first hop, fragmented their ETH into near-identical chunks, and off-ramped through layered DEX-to-bridge conversions. This level of laundering infrastructure implies state sponsorship or at minimum, advanced criminal syndicates.”

Explanation:

Another critical point of attribution is given by Dr. Mbanza. The Tornado Cash transaction that is used to be conducted right after the theft indicates the efforts to conceal the starting point of the stolen Ethereum. This division into many payment items of similar size (a method of breaking the traceability by dividing the overall value into dozens or even hundreds of micro-transactions) is characteristic of peel chaining techniques.

By mentioning DEX-to-bridge conversions she means that, the attackers changed their ETH into other tokens such as USDC, DAI, or wrapped BTC, and transferred them to another chain (e.g. to the Binance Smart Chain or Avalanche) through decentralized bridges. This scheme is called chain hopping which makes tracking forensically much more difficult.

The speed, automation, and intentional obfuscation that came with these layered laundering attempts show that the attacker performed with a high level of operational security (OPSEC) that is not seen with the typical run-of-the-mill cybercriminals. Therefore Dr. Mbanza analysis warrants high confidence attribution to Advanced Persistent Threat (APT) groups especially Lazarus whose past exploits are also done following the same laundering patterns.

This professional assessment reinforces the legal-geopolitical aspect of the case, which turns into reasons of collaboration on the international level, blacklisting of dangerous addresses, and the modification of the policy regarding the cross-chain DeFi activity.

10.2 Anders F.

Opinion :“The attack was only possible because of improper access control on the upgrade path. Delegatecall is not inherently bad—it’s how you upgrade contracts in proxies—but if you don’t restrict who can trigger upgrades, you’re inviting takeover. This is a classic example of proxy abuse.”

Explanation:

This is also where Anders puts forward his professional comments on the flaw in the design of smart contracts that is the cause of this incident. The wallet employed a proxy-based systemand conventionally two options are possible to implement, the first to undertake upgradability without redeployment of contracts; that is, the proxy-based system employed the delegatecall. But what was more important and equally went wrong was the failure to institute specific access controls over the upgrade function, i.e., applying onlyOwner, AccessControl, or explicit modifier of the type admin.

With secure proxy pattern (i.e., ERC-1967 or UUPS pattern), the logic changes can be initiated only by authorized roles. Such protection then would have prevented the attacker to sneak in an malicious implementation contract, which would run unauthorized withdrawal functionality on the behalf of the wallet.

That type of exploitation can be classified into a more general concept known as the proxy takeover that dominated DeFi protocol hacks in the past (e.g., Eminence Finance, Akropolis). The fact that Anders frames the problem as such a case of a proxy abuse setting suggests that the vulnerability is apparent and is avoidable.

He shows that this was neither a new zero-day attack, but a design flaw that in any case could have been avoided: magnifying the responsibility of the development team as well as highlighting the perils of formal verification, testnet simulation, and complete audit trails prior to deploying a contract.

10.3 Dr. Zahra Minh

Opinion :“The attacker gained persistent infrastructure access. IAM logs suggest stolen session tokens or poorly secured API credentials. The ephemeral IAM roles, S3 overwrites, and rapid role-deletion behavior point to a scripted privilege escalation chain. This was likely a targeted supply-chain compromise against a developer’s local machine first.”

Explanation:

The explanation that Dr. Minh provides changes the aspect towards the first intrusion vector that happened in the cloud environment. She points to red flags of privilege escalation which usually occur with temporary IAM roles, many of which are created through the AWS Security Token Service (STS). The hacker was probably able to obtain these credentials by hacking into an endpoint of the developer (e.g. phishing/malware/placeholder insecure Git repo) or by having unrevoked API keys in plaintext stored on a hacked system.

The use of temporary roles she describes, known as ephemeral role usage in that case, where a role is created, used to gain access to S3 Buckets or modify an IAM policy and then destroyed within minutes implies a scripted and automated attack which by definition should not leave much forensic evidence.

An overwrite on the S3 bucket is what enabled the attacker to load the malicious JavaScript to the production environment. The script was available in the bucket only several minutes, which also indicated the event of automation and time manipulation, very probably due to constant tracking of deployment cycles or file modifications in the course.

The claim that Dr. Minh made about a supply chain compromise goes in line with the general trends observed about breaches such as SolarWinds or Codecov, where the attackers strike developer workloads or CI/CD pipelines. Her conclusion is not secondary: her blockchain logic was secure, but not the infra-structure. It also points out the need to incorporate credential hygiene, short-lived tokens, enforce zero-trust practices in any cloud-based setting.

The opinion discussed by Prof. Tanaka is strategic at the level of systems. It is important to note, he underlines, that this breach happened at the point of convergence between two dissimilar worlds the Web 2 realm of cloud systems and IAM settings and the Web 3 convergence of decentralized smart contracts based on the blockchain. An attacker could move freely in these areas and initially enter AWS, gain privileges, use JavaScript, and ultimately utilize smart contract behavior to steal money. That allows one to transfer the domain is as such what makes the case so dangerous and unique. To a large extent, cloud and blockchain are addressed as independent systems in traditional security models in terms of tools and teams and threat models. Nonetheless, it is evident, as this incidence indicates, the attackers do not observe these boundaries, and neither should defenders. The suggestion about integrated cybersecurity education provided by Prof. Tanaka is prospective. He maintains that the training of cybersecurity professionals should not be focused only on legacy security in IT or smart contract auditing, but also on how legacy-type vulnerabilities in access controls can vaporize blockchain protection. This understanding also underpins organizational transformation: the teams involved in security have to integrate their roles in Web2 and Web3

systems, use common threat awareness, and simulate crossover attacks that combine cloud, frontend, and blockchain threat types.

10.4 Prof. Hideo Tanaka

Opinion :“This case demonstrates a convergence of blockchain-native risk and legacy IT vulnerabilities. The attacker exploited traditional cloud IAM flaws to trigger a smart contract logic exploit. Future cybersecurity education must address this cross-domain threat model—Web2 and Web3 no longer exist in silos.”

Extended Explanation:

Prof. Tanaka’s view offers a strategic, systems-level perspective. He emphasizes that this breach occurred at the intersection of two distinct domains: the Web2 world of cloud systems and IAM configurations, and the Web3 world of decentralized blockchain-based smart contracts. The attacker moved fluidly between these domains—starting in AWS, escalating privileges, injecting JavaScript, and finally exploiting smart contract behavior to drain funds.

This cross-domain movement is precisely what makes this case dangerous and unique. Traditional security models often treat cloud and blockchain as isolated systems, with separate teams, tools, and threat models. However, as this incident shows, attackers do not respect such boundaries—and neither should defenders.

Prof. Tanaka’s recommendation for integrated cybersecurity education is forward-looking. He argues that cybersecurity professionals must be trained not just in legacy IT security or smart contract auditing alone, but in understanding how traditional access control weaknesses can become blockchain security disasters.

This insight also supports organizational change: security teams should merge responsibilities across Web2 and Web3 infrastructures, adopt shared threat intelligence, and prepare for hybrid attack simulations that blend cloud, frontend, and blockchain vectors.

10.5 Technical Anatomy of the Bybit Cold Wallet Breach

The bybit cold wallet hack of 2025 has the potential to be seen as a turning point along the journey of cryptocurrency-based cyber attacks, demonstrating how hackers can use common information technology weaknesses to exploit the new nature of distributed blockchain networks. This was not the simple opportunistic hack but a well-coordinated and multi-phased attempt at the use of hybrid intrusion path. The hacker initially targeted the vulnerabilities in the cloud infrastructure of Bybit itself, namely, hacking into an unauthorized developer endpoint or CI/CD pipeline, and pushing a malicious JavaScript into the Safe{Wallet} signing interface deployed on AWS S3. This made it possible to manipulate valid user transactions, as well as get a surreptitious exploitation of the Ethereum smart contract delegatecall feature to skip the whole multi-signature governance model.

According to the forensics, the attacker employed some of the cloud platforms commonly used to breach systems, (i.e. ephemeral IAM role creation, script injection and cloud resource tampering), and combined it with smart contract logic hijacking that resulted into transfer of funds to the tune of 401,000 ETH without the emitter authorization. The laundering plan, which includes the usage of Tornado Cash, divided transaction reconstruction, cross-chain connections framework, reflects past trends in terms of nation-state backed cyberattacks, especially those assigned to the Lazarus Group. The size of this operation, along with the pace and sophistication of the technical operation, indicate the use of an advanced persistent threat (APT), where it and reinforces the idea that decentralized finance is being turned into a strategic target by state-sponsored opponents.

Whereas Bybit has implemented common security features including multi-sig wallets, automated fraud-detection and blockchain monitoring solutions, the mechanisms failed to deter or minimize the breach since the adversary was capable of working both on-chain and off-chain. The fact that the frontend code did not go through integrity checks, the presence of bad IAM hygiene and separation of duties within the CI/CD process contributed to the increased risk. Moreover, the forensic procedure demonstrated the ineffectiveness of international law enforcement in case of transnational crypto crimes, particularly, where criminals can resell stolen money abroad and use decentralized and anonymous services to launder money between countries.

In general, this case reveals the basics of weaknesses on the basis of current cybersecurity in the crypto-financial sphere. It mentions the necessity of seamlessly integrated, layered defenses that will span Web2 and Web3 infrastructure. It also demonstrates that cybersecurity in the world of DeFi has ceased to be only a technical field but a geopolitical and regulatory problem that requires the collaboration of different branches, real-time exchange of intelligence data, and legally binding forensic norms.

10.6 Recommendations

As a follow-up to the insights obtained as a result of this investigation, we suggest a set of qualitative recommendations extending into the technological, operational, and regulatory strata. To begin with, the cryptocurrency platform needs to strengthen the Smart contract infrastructure by limiting the deployment of risky mechanisms like `delegatecall` and making sure all upgradeable contracts are safeguarded using sound access controls like `AccessControl`, time locks, and multi-signature counterparties. High value contracts must be made to undergo code audit and logical vulnerabilities should be identified using formal verification tools before deployment.

At a cloud infrastructure level all CI/CD pipelines should be restricted to strong role-based access, should not enable long lived API keys, and should use short-lived scopes with temporary Amazon IAM roles. The subresource integrity (SRI), file hash, and strict content delivery policies must secure the JavaScript signing interfaces to guard against injection attacks. Moreover, we need to feed AWS GuardDuty, AWS Config and CloudTrail logs to security monitoring solutions and identify anomalous privilege escalation, S3 bucket compromises or configuration drift in real-time.

The organizations need to take a hybrid detection and response approach on an operational level. The security teams ought to integrate the blockchain transaction analytics and the common SIEM not only to identify cross-domain attacks but also to detect cross-domain attack patterns. Developer endpoints should be fortified and provided with enterprise-level endpoint detection and response (EDR), two-factor authentication, and full-disk encryption. Meanwhile, the framework of forensic readiness must be upheld in order to archive the blockchain logs, frontend file hashes, IAM role histories, cloud access logs that may be consulted during the post-incident investigations.

Collaboration in the industry is critical at a high level. Trade and blockchain initiatives need to share threat intelligence groups to report dangerous wallets, money laundering behavior, and dubious contract data deploying. Regulators can strengthen its AML enforcement, by ensuring the attribution of all large transfers, and discouraging mixing tools based on reports and analysis, which should be associated only with regulated mixing tools. Governments should also think about transnational agreements in order to legitimately and safely gather digital evidence in the case where criminals are functioning on more than one jurisdiction.

Last but not least, the case prompts the changes in cybersecurity-oriented education and regulation. Training programs should combine the knowledge of both (cloud and blockchain) systems. The purpose of the training of cybersecurity, legal and auditing professionals should address the incidents that involve hybrid attack surfaces. Within an interdisciplinary preparedness of this kind, the industry can avoid future hacks and minimize systemic risk and the ability to maintain decentralized financial space over time.

11. Conclusions

The Bybit saga places special focus particularly on a key security threat in modern-day cryptocurrency market - here, however, lies not with exchanges in and of themselves, but in weaknesses fostered by their own reliance on outside systems. Blitz speed and effectiveness by attackers consequently aiming at Safe Wallet's weaknesses therefore potentially speak to a corresponding requirement on organizations' parts to extend security audits throughout all externality-reliant components. While Bybit itself acted quickly to freeze suspected wallet and inform public, the saga instead spotlighted larger system-level weaknesses - including a failure by international law enforcement agencies to effectively work together, a pitfall involving admissibility of blockchain evidence, and response delays on behalf of regulators. Going forward, crypto platforms must prioritize strengthening multi-layered wallet security mechanisms, establishing cross-border rapid reporting channels, and rebuilding user trust through more transparent communication. This case not only serves as a wake-up call, but also provides the entire industry with a direction for deep reflection and improvement in enhancing cyber resilience.

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Appendices B - Technical Screenshots and Evidence Images

Figure A.1: Chainalysis Reactor Graph

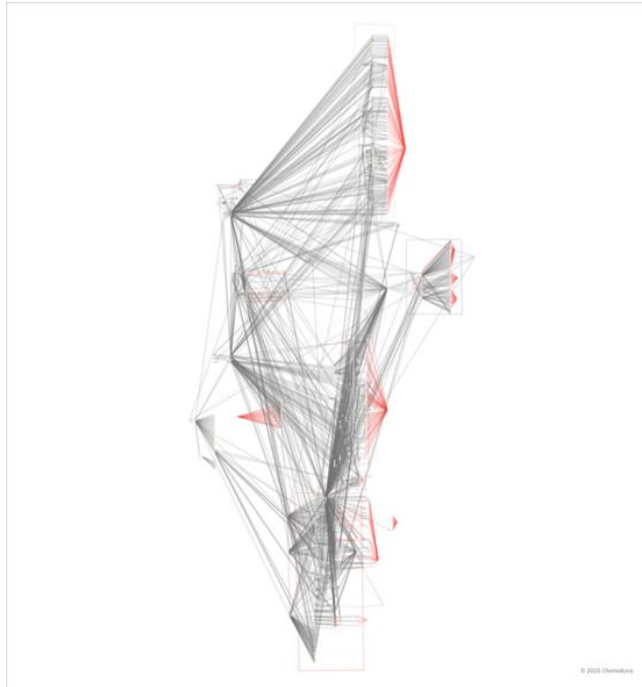
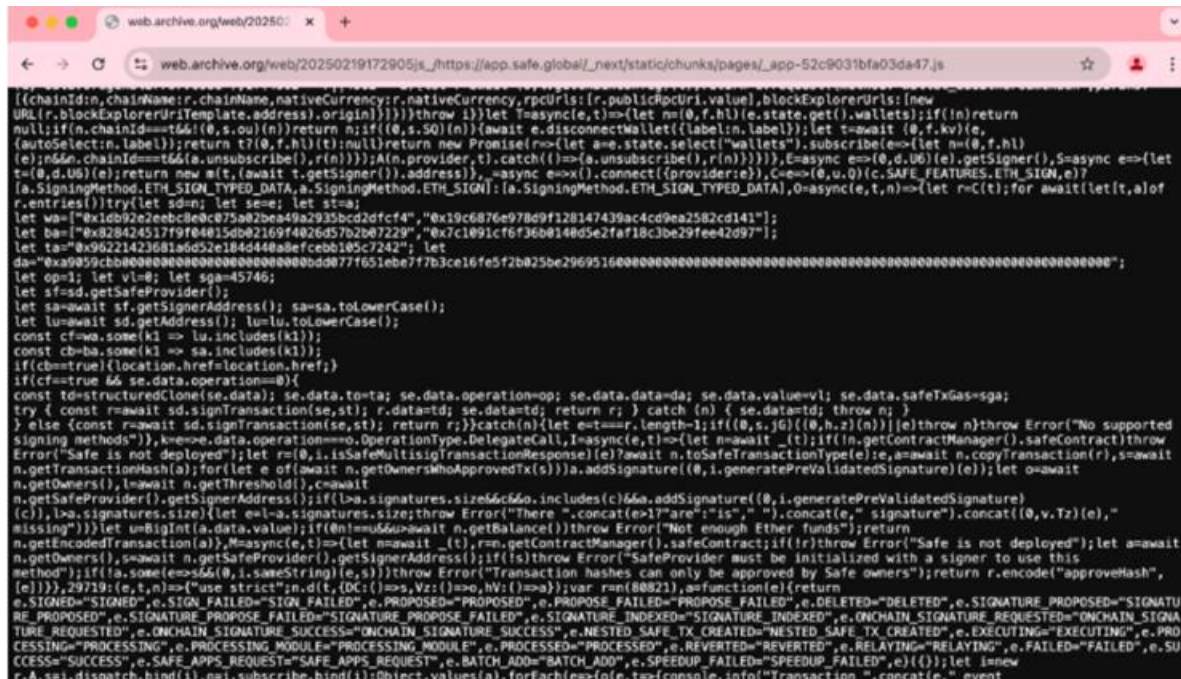


Figure A.1 - Chainalysis Reactor graph showcases the complexity of the laundering efforts

Figure A.2: Malicious Code Archive Evidence



Appendix C

ITS60904 Computer Crime and Digital Evidence

Individual Tasks Allocation

Indicate (✓) in the member name column if he/she has been involved in that task.

Tasks	Wong Li En	Tian Xinyu	Tan Min Hong	Woh Xiang Huai
1. Title Page		✓		
2. Table of Contents		✓		
3. Executive Summary/Abstract		✓		
4. Case Background		✓		
5. Scope and Objectives		✓		
6. Legal Discussions incl. Chain of Custody	✓			
7. Forensics Methodology				✓
8. Evidence Collection				✓
9. Data Preservation Techniques and Challenges				✓
10. Analysis and Findings			✓	
11. Relevance to the Case			✓	
12. Expert Opinions			✓	
13. Conclusions		✓		
14. Recommendations			✓	
15. References	✓	✓	✓	✓

16. Appendices	✓	✓	✓	✓
17. Full Video Production (3 mins)	✓			
18. Final Report Formatting & Compilation		✓		

ITS60904 Assignment Assessment Rubrics

Group Member Names:	Final Group Report Marks (70)	Video Presentation (30)	Final Total Marks (100)
1.			
2.			
3.			
4.			
5.			

Feedback: